

Performance Testing Report

Application

AI Dashboard Web Application (Sample Application)

Objective

Verify application performance under normal operating conditions.

Test Environment

Parameter	Value
Browser	Chrome 138
OS	Windows 11
Network	100 Mbps
Test Tool	JMeter
Test Duration	30 Minutes

Performance Test Cases

TC ID	Scenario	Expected Result	Actual Result	Status
PT-001	Load Login Page	< 3 sec	2.1 sec	Pass
PT-002	User Login	< 2 sec	1.4 sec	Pass
PT-003	Dashboard Load	< 5 sec	3.8 sec	Pass
PT-004	Search Records	< 3 sec	2.5 sec	Pass
PT-005	Generate Report	< 8 sec	6.3 sec	Pass
PT-006	Export Data	< 10 sec	7.1 sec	Pass
PT-007	Upload File	< 5 sec	4.2 sec	Pass
PT-008	API Response Time	< 1 sec	0.6 sec	Pass
PT-009	Database Query	< 2 sec	1.2 sec	Pass
PT-010	Logout Process	< 2 sec	0.8 sec	Pass

Load Testing Execution

Scenario 1: 50 Concurrent Users

Metric	Result
Virtual Users	50
Average Response Time	1.8 sec
Throughput	250 req/min
Error Rate	0%
CPU Usage	45%
Memory Usage	55%

Result

System Stable

Scenario 2: 100 Concurrent Users

Metric	Result
Virtual Users	100
Average Response Time	2.9 sec
Throughput	480 req/min
Error Rate	1%
CPU Usage	68%
Memory Usage	72%

Result

Acceptable Performance

Scenario 3: 500 Concurrent Users

Metric	Result
Virtual Users	500
Average Response Time	7.4 sec
Throughput	1900 req/min
Error Rate	8%
CPU Usage	92%
Memory Usage	95%

Result: Performance Degradation Observed

Load Testing Summary

Metric	Result
Maximum Users Tested	500
Peak Throughput	1900 req/min
Average Response Time	4.03 sec
Highest CPU Usage	92%
Highest Memory Usage	95%
Error Rate	8%